Investigating the migration velocity of UV-blocking chemical compounds in novel photovoltaic module topologies

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I. Introduction

A PV module is composed of a stack of active and protective materials. At the heart of this device are the **interconnected cells** that convert light into electricity. They are packaged with a **polymeric encapsulant**, **glass** on the front, and a **backsheet** or glass at the rear side. An **edge sealant** can be added in order to inhibit moisture ingress. This architecture is necessary to ensure protection from external factors and maintain the structural integrity of the cells. However, when exposed to **ultraviolet radiation** (UVs for short), PV modules are subject to delamination, encapsulant loss of adhesion or discoloration, and ground faults due to backsheet degradation [1]. These phenomena can be mitigated with the addition of **UV blockers** in the encapsulant. In this situation, it is preferable that only the rear encapsulant contain UV blockers, as the module performance would be decreased by a UV blocking front encapsulant (before the cell). Nevertheless, some cell technologies degrade under UVs, and front UV protection is therefore necessary. In such cases, **UV downshifters** are an alternative to UV blockers. These are materials that absorb UVs and re-emit photons with lower energy. This strategy allows for UV protection while reducing performance losses compared to UV blockers.

Both strategies have been shown to prevent or mitigate UV degradation, but in preliminary work, it was also observed that UV blocking and downshifting molecules can **diffuse in the module through the encapsulant**. This phenomenon occurs during field ageing as well as **accelerated ageing tests**, and has been reported in literature but not studied in detail [2] [3]. More generally, additive migration between polymeric materials of PV modules has been observed, for example from the backsheet to the encapsulant [4].

In the literature, factors influencing the **diffusion coefficient** of additives are known, such as the nature of the diffusing molecule, its molecular weight and geometry, in addition to the polymer matrix through its structure and mobility. External factors like heat and humidity that result in material swelling increase the diffusion coefficient. The goal of this work is therefore to understand the phenomena that are responsible for the observed migration, focusing on the role of **external factors (heat, moisture, UVs)** but also on the nature of both polymers and additives. Moreover, as this effect should be avoided, the objective is to **minimise the diffusion** through the choice of the

right base polymer and UV blocker. The experimental work will consist of investigating the speed of migration of UV blocking and downshifting molecules in the polymeric encapsulant with various types of polymeric resins, UV blocking molecules, and ageing tests.

Group formation & Referee

Both group members are part of the EPFL PV-LAB Module and Reliability Group, based in Neuchâtel. The proposed project is closely related to Lison's work within CSEM and PV-LAB, and will involve collaboration with her thesis co-director, Dr. Hengyu Li, who has been selected as the project referee.

II. Project description

A. Mind-map

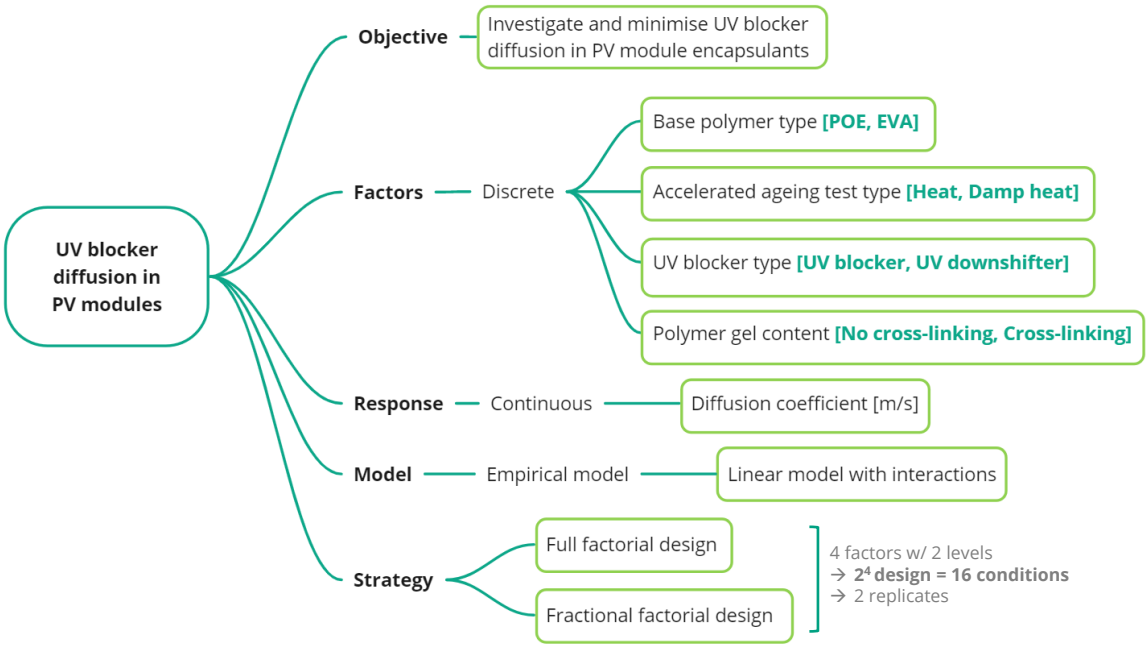

Fig. 1: Project Mind-map.

B. Objective

The objective of the project is to determine the main factors and interactions affecting the diffusion coefficient of UV blocking molecules in polymeric encapsulants, and the results will enable to minimise this undesirable effect.

C. Factors

The factors that will be evaluated are the following:

- **Base polymer type** | Discrete | 2 levels [Polyolefin Elastomer (POE), Ethylene Vinyl Acetate (EVA)]
- **Accelerated ageing test type** | Discrete | 2 levels [Heat, Damp Heat]
- **UV blocker type** | Discrete | 2 levels [UV blocker, UV downshifter]
- **Polymer gel content** | Discrete | 2 levels [No cross-linking, Cross-linking]

Regarding the choice of base resins, EVA is selected as it is used in 90% of today's production, whilst POE is increasingly used in novel PV module topologies. The created sample modules will be aged in different climatic chambers according to PV degradation norms for the accelerated ageing test. To account for the various molecular compounds that can be used for UV blocking effects, two

UV blocker types and one UV downshifter are chosen. Finally, the polymer gel content describes the cross-linking between polymers and is simplified to two levels for the purpose of this experiment: the presence or absence of cross-linking.

D. Response

The responses that will be reported is the **diffusion coefficient [m/s]** of the UV blocking molecules (see Fig. 2).

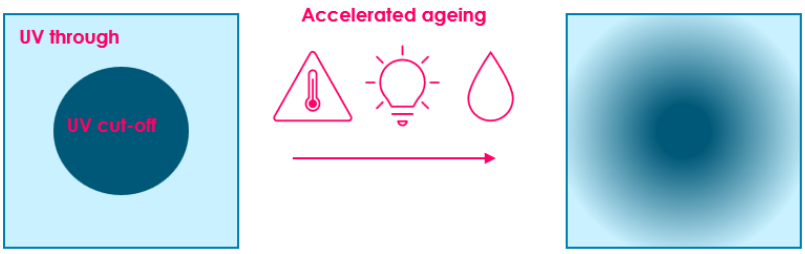

Fig. 2: Schema of the samples before and after ageing.

E. Model

We will use pure empirical model, 1st degree and 1st degree with interactions will be evaluated.

F. Strategies

We have decided to work on a full factorial design with two replicates per condition. This method allows to proceed an empirical linear model with interactions. Full factorial design in our experiment represents 2^4=16 conditions. We also would like to determine if similar conclusions are achieved through fractional factorial design, which represents 2^{4-1} experiments. We are also aware that the main effects will be aliased with the three way interactions and that the two way interactions are aliased with others. Therefore the fractional factorial design will only allow to estimate the four main effects.

G. Program

Fig. 3 shows the preliminary project schedule in the form of a Gantt chart, highlighting the main work packages (WP) and experimental steps.

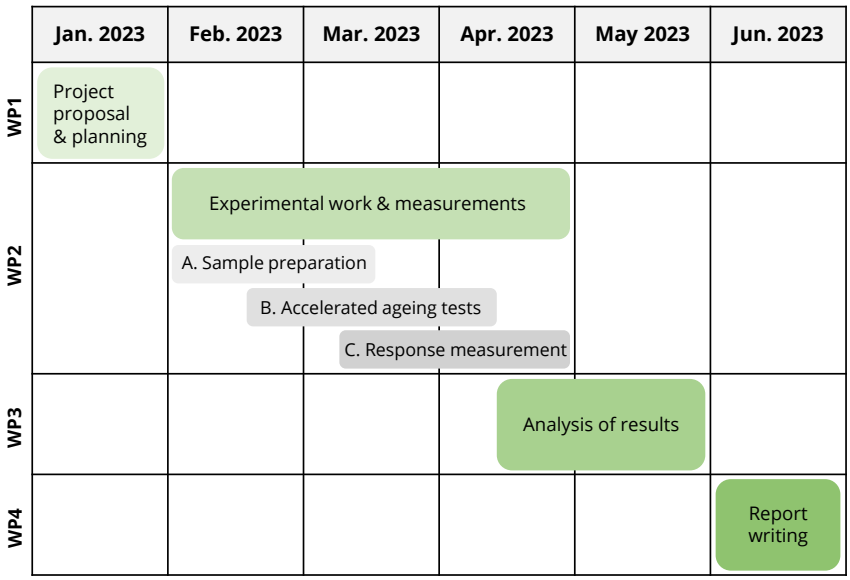

Fig. 3: Project Gantt chart and work packages (WP).

III. Remarks

A. *Confidentiality*

This project deals with confidential data and information proprietary to CSEM. Therefore, this report and work should not be shared with any third party.

References

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